

CHALLIS, ID, USA

WMO: 722142

Lat: 44.523N

Lon: 114.215W

Elev: 5072

StdP: 12.19

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 04114

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-2.2	-14.9	3.0	-7.0	-9.2	4.0	5.9	17.5	33.2	15.3	33.5	0.9	320	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.4	91.9	57.7	89.8	57.1	86.4	56.5	61.2	82.4	59.7	81.3	58.3	80.2	10.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.5	76.5	66.0	52.1	69.9	65.0	49.6	63.4	63.9	29.5	82.2	28.3	80.5	27.3	80.8	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	16.8	14.0	DB	-15.6	97.2	6.6	2.5	-20.3	99.0	-24.2	100.5	-27.9	101.9	-32.7	103.7	(4)
(5)			WB	-15.9	64.9	6.3	1.7	-20.4	66.1	-24.0	67.1	-27.6	68.0	-32.1	69.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	19.8	27.6	36.9	43.8	52.8	61.2	70.0	66.9	58.2	44.6	31.7	19.9	(6)	
	DBStd	18.66	10.99	9.50	6.94	7.02	7.05	7.03	4.69	4.71	6.87	7.36	9.18	10.63	(7)	
	HDD50	3937	936	629	407	208	52	6	0	0	17	197	550	935	(8)	
	HDD65	7785	1401	1049	871	637	380	151	14	34	216	633	999	1400	(9)	
	CDD50	1940	0	0	1	21	139	342	619	525	263	29	1	0	(10)	
	CDD65	313	0	0	0	0	2	37	168	94	12	0	0	0	(11)	
	CDH74	6298	0	0	0	10	149	938	2814	1861	513	13	0	0	(12)	
	CDH80	2564	0	0	0	0	21	328	1325	755	134	1	0	0	(13)	
(14) Wind	WSAvg	4.2	2.3	3.5	4.8	5.9	5.6	5.5	5.1	4.6	3.9	3.5	3.0	2.3	(14)	
(15) Precipitation	PrecAvg	7.8	0.5	0.4	0.5	0.6	1.1	1.1	0.7	0.6	0.7	0.4	0.6	0.6	(15)	
	PrecMax	12.0	1.7	1.3	1.7	1.7	3.1	3.8	2.6	1.9	2.3	1.8	1.7	3.7	(16)	
	PrecMin	4.8	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	1.8	0.4	0.3	0.4	0.4	0.7	0.8	0.6	0.5	0.7	0.4	0.4	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	53.7	64.8	74.9	82.3	91.8	96.9	93.4	88.4	74.8	58.1	47.9	(19)	
		MCWB	36.9	42.0	44.7	49.5	53.9	57.8	59.6	57.8	56.6	50.9	46.1	39.8	(20)	
	2%	DB	40.9	48.2	59.3	69.6	78.7	87.9	93.3	90.4	83.8	68.5	53.9	43.0	(21)	
		MCWB	34.1	38.1	42.2	47.3	52.1	56.4	58.2	57.0	54.5	48.8	42.3	36.4	(22)	
	5%	DB	37.1	44.6	54.4	64.1	73.8	83.7	90.8	88.0	80.7	63.4	48.8	38.7	(23)	
		MCWB	32.0	35.9	40.4	44.3	50.7	55.0	57.5	56.7	53.6	47.1	40.1	33.6	(24)	
	10%	DB	34.2	40.7	50.2	59.0	69.4	80.0	88.2	84.2	75.5	58.6	45.3	34.5	(25)	
		MCWB	30.3	33.8	38.6	42.8	49.2	54.1	57.1	55.8	52.0	45.1	38.1	30.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	43.2	46.4	50.8	57.0	61.3	63.5	62.5	59.8	53.4	47.5	40.5	(27)	
		MCDB	43.8	51.9	60.9	72.5	72.9	84.0	83.6	82.2	77.9	68.5	56.4	46.7	(28)	
	2%	WB	35.3	39.4	43.7	47.6	54.1	58.9	61.6	60.4	57.3	50.8	43.5	37.4	(29)	
		MCDB	39.9	46.1	55.5	65.5	72.4	80.9	83.1	80.8	77.1	64.2	51.4	41.7	(30)	
	5%	WB	32.9	36.9	41.6	45.5	52.0	57.1	60.2	58.8	55.3	48.3	41.1	34.0	(31)	
		MCDB	36.5	42.9	52.5	61.6	70.3	77.8	82.4	79.6	75.2	61.0	47.9	38.2	(32)	
	10%	WB	30.4	34.4	39.4	43.6	50.2	55.3	58.9	57.3	53.3	46.0	38.4	31.0	(33)	
		MCDB	33.8	40.0	48.9	57.5	66.8	75.6	81.6	79.2	72.3	57.7	44.4	34.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	19.7	23.1	25.0	25.9	29.4	34.4	34.3	31.8	25.7	20.4	17.8	(35)	
		MCDBR	19.8	22.5	30.1	34.4	34.4	37.7	38.7	39.4	38.7	32.8	24.8	17.9	(36)	
	5% WB	MCWBR	14.4	14.2	16.9	16.8	15.2	14.6	13.9	15.8	17.2	16.9	15.3	13.0	(37)	
		MCWBR	18.1	20.6	27.0	30.3	31.0	33.4	34.2	34.5	34.7	29.8	22.7	16.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.245	0.254	0.274	0.327	0.322	0.320	0.329	0.332	0.307	0.291	0.263	0.247	(40)	
		taud	2.395	2.365	2.423	2.354	2.456	2.504	2.505	2.465	2.542	2.549	2.546	2.449	(41)	
	Ebn at Noon		285	301	308	297	300	300	296	292	293	285	278	273	(42)	
		Edh at Noon	23	29	31	37	35	33	33	33	28	24	20	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		567	885	1305	1666	1889	2178	2268	1961	1540	1005	601	442	(44)	
	RadStd		60	93	96	116	183	174	158	116	127	82	52	49	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	N/A	
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	N/A	

Nomenclature: See separate page